

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Eighth Semester B.E. (ISE) (Credit System) Degree Examinations

May- June 2023

19ISE43 – SOFTWARE ARCHITECTURE

Duration: 3 Hours

Max. Marks: 100

Note Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Define Software Process. Explain the Architectural Activities in Software Architecture.	10	L*2	1	1,2
	b)	Explain: i) Architectural Pattern ii) Reference Model iii) Reference Architecture iv) Bring out the difference between the above with neat diagram.	10	L2	1	1,2
2.	a)	Explain the Common Software Architectural Structures in detail.	10	L2	1	1,2
	b)	What are the 3 issues involved in System Quality Attributes and explain the solution for System Quality Attributes with neat diagram.	10	L2	1	1,2
3.	a)	Define Software Testability. Explain the testability Scenario with a neat diagram.	10	L2	2	1,2
	b)	Explain the Fault Recovery and Fault Prevention tactics in detail.	10	L2	2	1,2
		Unit – II				
4.	a)	Explain Layered Architectural Style with a neat diagram.	10	L2	2	1,2
	b)	Explain Pipe and Filter Architectural Style with neat diagram.	10	L2	3	1,2
5.	a)	Explain Event Based Architectural Style with a neat diagram.	10	L2	3	1,2
	b)	Explain MVC Architectural Pattern with diagram.	10	L2	3	1,2
6.	a)	Explain how to implement a PAC Architecture.	10	L2	2	1,2
	b)	Explain the variants of broker architectural Pattern.	10	L2	2	1,2
		Unit – III				
7.	a)	Explain Evolutionary Life Cycle in detail.	10	L2	4	3
	b)	Explain attribute driven design and identify the steps involved in attribute driven design.	10	L2	4	3
8.	a)	Explain the documenting a view with diagram.	10	L2	5	1,2
	b)	Explain the Cross View Documentation.	10	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
